Yahya Jabary

jabaryyahya@gmail.com | Swiss B-Permit, EU Citizen (Austrian) | Native German, English fluent

+41 78 221 22 60 | 8052 Zurich, Switzerland

GitHub github.com/sueszli (70+ projects)

Google Scholar scholar.google.com/citations?user=SH4-xwEAAAAJ

Linkedin linkedin.com/in/sueszli

Education

ETH Zurich, Master's Thesis · *Machine Learning* — *Jul 2024 - Mar 2025*

TU Wien, Master's Degree · *Computer Science* — *Oct 2019 - Mar 2025*

Experience

University of Cambridge — Remote / Cambridge, UK

Research Assistant in Machine Learning Compilers — *Nov 2025 - present*

- Reimplemented MicroGPT on a custom vectorized JIT compiler with a user-scheduled frontend.
- Hit 5.3x speedup over PyTorch and 14.4x over JAX via kernel fusion and autovectorization.
- Prototyped an Apple GPU backend from reverse-engineered ISA notes.

Python C++ MLIR LLVM

Qube Research & Technologies — Paris, France

Quantitative Technology Analyst, Contract — *Apr 2025 - Oct 2025*

- Designed an ensemble of statistical volatility indicators driving intraday signals.
- Ported hot paths to zero-copy CPython extensions in Rust via PyO3 for 2.1x speedup.
- Patched upstream FFI alignment faults at the Python/Rust boundary in Arrow's internals.
- Built a differential fuzzing harness for the port that caught 100+ numerical regressions.

Python Rust Apache Arrow

ETH Zurich — Zurich, Switzerland

Research Assistant in Machine Learning Robustness — *Jul 2024 - Mar 2025*

- Disproved DeepMind's "self-ensembling robustness" paper via full reproduction and PGD attacks.
- First-authored paper on a CAPTCHA-based Out-Of-Distribution eval suite for vision models.
- Follow-up by co-authors accepted at NeurIPS 2025.
- Cut experiment turnaround 5x via distillation, distributed training and post-training quantization.

Letter of Recommendation: sueszli.github.io/rec.pdf

Python PyTorch Singularity/Apptainer

University of Waterloo — Ontario, Canada

Research Assistant in Operating Systems — *Nov 2023 - Jun 2024*

- Offered fast-tracked PhD directly from my Bachelor's. Declined.
- Ran the first large-scale vector-DB benchmark for code retrieval, with queries synthesized via AST mutation.
- Virtualized an LLVM-syscall record-and-replay for Golang in FreeBSD jails for reproducible runs.

C Python Faiss Lucene

Erste Asset Management — Vienna, Austria

Trading Software Engineer, Contract — *Apr 2023 - Oct 2023*

- Replaced per-message subprocess spawns with an in-process pipeline, cutting tick-to-signal latency.
- Reported a boot-order privilege escalation bypassing endpoint security to the CISO.

Java Python Kafka FIX Protocol

Publications

Jabary, Y. (2025). Rethinking Adversarial Examples (Master's Thesis, ETH Zurich). [code, pdf]

Jabary, Y., Plesner, A., Kuzhagaliyev, T., & Wattenhofer, R. (2024). Seeing Through the Mask: Rethinking Adversarial Examples for CAPTCHAs. [press, code, pdf]

Awards & Honors

Qube Research & Technologies / Raspberry Pi Foundation, Nominated Contest Judge	*2025*
Google for Developers, Nominated Community Chapter Leader	*2024*
EPFL, Summer Research Institute SuRI Fellowship	*2024*
University of Waterloo, Mathematics International Master's Award of Excellence M-IMAE ($12,500)	*2023*
University of Waterloo, Graduate Research Studentship GRS ($23,081)	*2023*
TU Wien, Institute of Logic & Computation Research Scholarship (€1,200)	*2021*
Austrian Government, National Maker Competition Prize (€3,000)	*2018*
University of Vienna, Early Admission Program in Mathematics for Gifted Students	*2017*